Selected bidiagonal singular directions with fixed-format linear output work

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Abstract

Computing a few singular vectors of a bidiagonal matrix requires control of orthogonality as well as residuals when singular values cluster. We give a rational-filter construction for any requested set of k ordered singular triplets of a real $n \times n$ upper bidiagonal matrix. For $\epsilon = 2^{-p}$, $16 \le p \le 53$ and $n\epsilon \le 1/64$, its represented unit directions satisfy both singular-vector residual bounds, ordered-value accuracy and separate left and right orthogonality bounds with a universal constant, independent of the input gaps. The work is $O(p^4nk)$ operations on $O(p)$-bit numbers, or conservatively $O(p^6nk)$ bit operations. Thus the work is linear in nk after the format is fixed. The sufficient core precision is 220224 bits for a binary64 target; practical speed superiority is not claimed. Count-separated spectral intervals, positive rational filters and joint Schur-complement columns avoid a final vector reorthogonalization. A common cut near zero permits recovery of left directions without a small-singular-value assumption. The arithmetic model and the distinction between floating carriers and exact unit directions are explicit. A research implementation supplies separate proof and diagnostic configurations with optional exact checks of finite outputs.

1 Introduction

The singular value decomposition of a dense matrix is commonly reduced to the same problem for a bidiagonal matrix. If only a few singular triplets are needed, the final bidiagonal stage should ideally require work proportional to the number of returned entries. Small residuals alone do not ensure this: vectors associated with a tight cluster can satisfy their individual equations while being almost parallel. Selected triplets must satisfy both singular-vector equations and must remain separately orthogonal on the left and on the right.

Großer and Lang [2] studied an $O(n^2)$ bidiagonal SVD construction using relatively robust representations. Their analysis identifies difficulties with separately solving the two Gram eigenproblems, and with tiny clusters in the Jordan–Wielandt formulation. Marques, Demmel and Vasconcelos [3] developed selected bidiagonal SVD computation through an associated symmetric tridiagonal eigenproblem. Their report discusses the care required in extracting singular vectors and the limitations of available clustered-eigenvector computations. These are different issues from obtaining a small individual eigenpair residual.

Problem 3.8 of Amsel et al. [1] asks for $O(nk)$ bidiagonal singular-vector work together with residual and orthogonality guarantees. We make the arithmetic and output conventions explicit. Our target additionally requires accuracy at each requested ordered singular value and both residual equations. The construction below has a universal error constant for these simultaneous requirements. Its $O(nk)$ statement concerns a fixed format: the precision dependence is $O(p^4nk)$ multiprecision

operations. We represent an exact unit direction by a nonzero floating carrier x with meaning $x/\sqrt{x^T x}$; the raw arrays satisfy a separate perturbation bound.

Informally, for any stored real upper bidiagonal input and any distinct requested ranks, we obtain nonnegative singular estimates and unit directions with value and residual errors at most $128n\epsilon \|B\|_2$ and pairwise inner products at most $128n\epsilon$. No separation between neighboring input singular values is assumed. The construction uses correctly rounded real operations with protected exact index and scaling arithmetic, excluding overflow and underflow. Its very conservative precision makes it a complexity construction rather than a proposed replacement for standard SVD software.

The first difficulty is to make the requested vectors joint rather than independent approximations. Protected inertia counts produce disjoint spectral intervals whose endpoints lie in gaps chosen by the algorithm. A positive rational approximation to each interval projector permits successive Schur-complement columns. A trace estimate chooses their coordinates while controlling a single orthogonal comparison family. Semiseparable resolvent representations make each coordinate scan linear in the matrix order.

The second difficulty is finite arithmetic in repeated shortening and recovery near zero. We compare the supplied generators with one nearby filter and bound scan errors over an entire segment, avoiding a product of local stability constants over the matrix order. For recovery, one common cut of the squared spectrum divides the small-value and positive branches. The near-kernel bound on a left direction cancels the apparent reciprocal small-value loss in its inner product with a positive left direction.

Section 2 gives the model and main result. Sections 3–5 construct and analyze the joint tridiagonal procedure. Section 6 proves the bidiagonal recovery and rounding bounds, and Section 7 describes the implementation and finite validation. Vector norms are Euclidean and matrix norms are spectral unless stated otherwise. A superscript T is an ordinary transpose, including in complex symmetric formulas; $*$ denotes an adjoint.

2 Task, arithmetic and output

Let $B \in \mathbb{R}^{n\times n}$ have diagonal $d_1, \dots, d_n$ and superdiagonal $e_1, \dots, e_{n-1}$. The input entries are the stored real floating numbers, not unknown numbers before input rounding. Write its singular values as $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The requested indices $j_1, \dots, j_k$ are distinct elements of $\{1, \dots, n\}$, with $1 \leq k \leq n$.

Fix $p \in \{16, \dots, 53\}$ and put $\epsilon = 2^{-p}$. The output carriers and singular estimates use binary64 storage; $p < 53$ requests a looser target for that same stored input. Basic real addition, subtraction, multiplication, division and square root at each specified internal precision are rounded to nearest. Complex arithmetic is expanded into these real operations. Protected integer indices, count-grid operations and power-of-two exponents are exact. Overflow and underflow are excluded. Input exponent costs belong to the fixed input format. The mathematical result is under this arithmetic model, rather than a formal verification of a language, compiler or library.

For each nonzero real carrier x, define

$$\mathcal{U}(x) = \frac{x}{\sqrt{x^T x}}. \tag{1}$$

An output $(\widehat{\sigma}, x, y)$ represents the exact mathematical triplet $(\widehat{\sigma}, \mathcal{U}(x), \mathcal{U}(y))$. Equation (1) specifies the direction represented by a finite array; it is not an assertion that an array has exactly unit floating norm. Constructing, copying and approximately materializing the directions is charged. An explicitly requested exact squared norm is a separately paid rational sum; no exact irrational floating primitive is assumed.

Theorem 2.1. *Suppose $n\epsilon \le 1/64$ under the preceding model. For every such B and every set of k distinct requested indices, the algorithm below returns nonnegative binary64 estimates $\widehat{\sigma}_i$ and nonzero binary64 carriers x_i, y_i. With $\widehat{u}_i = \mathcal{U}(x_i)$ and $\widehat{v}_i = \mathcal{U}(y_i)$,*

$$|\widehat{\sigma}_i - \sigma_{j_i}(B)| \le 128n\epsilon \, \|B\|, \tag{2}$$

$$\|B\widehat{v}_i - \widehat{\sigma}_i\widehat{u}_i\|, \quad \left\|B^T\widehat{u}_i - \widehat{\sigma}_i\widehat{v}_i\right\| \le 128n\epsilon \, \|B\|, \tag{3}$$

$$\left|\widehat{u}_i^T\widehat{u}_j\right|, \quad \left|\widehat{v}_i^T\widehat{v}_j\right| \le 128n\epsilon \quad (i \ne j). \tag{4}$$

Also $\left|x_i^T x_i - 1\right|, \left|y_i^T y_i - 1\right| \le 8\epsilon$. The constant is independent of all singular-value gaps and clusters. The complete production work is $O(p^4nk)$ operations on $O(p)$-bit numbers; elementary multiprecision arithmetic gives the conservative bit bound $O(p^6nk)$. The sufficient kernel precision is $67584 + 2880p$ bits, including 220224 bits when $p = 53$.

Here n is the matrix order, k the number of requested ranks and ϵ the target error scale. Repeated values permit any orthonormal directions in their subspace. Signed, reducible and singular matrices are included. Fixing the format makes the displayed operation bound $O(nk)$ in n, k; the theorem does not give a precision-uniform $O(nk)$ bound.

2.1 The bidiagonal procedure

For $B \ne 0$, let $\rho = \max(|d_i|, |e_i|)$ and choose a power of two s with $2\rho \le s \le 4\rho$. Set

$$A = B/s, \qquad G_R = A^TA, \quad G_L = AA^T, \qquad \Delta = n\epsilon, \quad \xi = 2^{-20}n\epsilon^3, \quad q = 3p + 32. \tag{5}$$

The row and column sums of a bidiagonal matrix give $1/4 \le \|A\| \le 1$ and $s \le 4\|B\|$. Both raw Grams are tridiagonal and satisfy $0 \le G_R, G_L \le I$. Their local products are formed from the stored input at the specified precision, not from a binary64 rounded Gram. They have the same ordered eigenvalues with multiplicity.

The tridiagonal procedure analyzed below returns jointly controlled directions and centers for any requested ranks of G. For accuracy $\eta = 2^{-q}$ its value, residual and pairwise inner-product errors are at most $512n\eta$. Call this procedure JointGram(G, J, q).

Algorithm 1: selected bidiagonal singular directions. Input: stored $d \in \mathbb{R}^n$, $e \in \mathbb{R}^{n-1}$, distinct ranks $J = (j_i)_{i=1}^k$, and p with $n2^{-p} \le 1/64$.

1. If $B = 0$, return the requested distinct coordinate vectors on both sides and zero values. Otherwise form (5).
2. Use protected counts of G_R to find a common cut $\theta \in [\Delta^2, 2\Delta^2]$ with clearance $\gamma > \Delta^2/(64n)$. Let m_0 be the exact number below the cut.
3. Call JointGram(G_R, J, q) once for all requested right directions v_i and centers α_i.
4. Call JointGram(G_L, J_0, q) once, only for requested indices $J_0 = \{j_i : j_i > n - m_0\}$, giving w_i, β_i.
5. For $j_i \notin J_0$, compute $t_i = \|Av_i\|$, $u_i = Av_i/t_i$ and $\widetilde{\sigma}_i = st_i$. For $j_i \in J_0$, set $u_i = w_i$ and $\widetilde{\sigma}_i = 0$.
6. Round values and carriers only after this recovery; return their unit representations (1) in requested order.

Output: k nonnegative values and k left/right carriers. Optional exact instance verification is outside this production algorithm.

The safe cut costs $O(n\log(n+1))$ work. Packet placement and localization cost $O(nk[q+\log(n+1)])$. With rational rank $r=O(q^2)$, coefficient construction costs $O(r)$ per packet, supply costs $O(nr)$, each selected scan $O(nr^2)$ and each selected solve/matvec $O(nr)$. There are at most $k+k_0\le 2k$ requested Gram columns. Since $\log n\le p$, these costs, recovery and output give $O(p^4nk)$. Storage is $O(nk+nr+r^2)$ high-precision words, with output, controller and one packet workspace counted. No full spectrum, $n\times n$ projector, entire null space, or final k-vector reorthogonalization is computed.

3 Protected counts and rational interval filters

We first separate spectral intervals without a promise on the input gaps. The spectral theorem for real symmetric matrices [4, Theorem 7.29] gives ordered eigenvalues and orthogonal projectors. Its min–max formula implies $|\lambda_j(T)-\lambda_j(H)|\le\|T-H\|$: each unit Rayleigh quotient changes by at most this norm, and one takes the same minima and maxima. Invertible congruence preserves inertia because it preserves the maximum dimension of a strictly negative quadratic subspace. These facts justify the protected counts that follow.

3.1 Counts and rank localization

Put $Q=2^{12q}$, $\tau=\eta^4$ and $\beta=\tau+4/Q$. Round each tridiagonal entry to the $1/Q$ grid. At a grid shift z evaluate the scalar LDL^T recurrence, rounding each quotient to this grid. Replace a pivot of magnitude below τ by the same-signed τ, assigning a zero pivot the negative sign. Count the negative pivots.

These pivots are exact factors of a nearby tridiagonal: the quotient rounding and pivot replacement alter a diagonal entry by at most $\tau+1/(2Q)$. Entry rounding contributes norm error below $2/Q$. Thus, if $N(z)=\#\{\lambda<z\}$ and $c(z)$ is the computed count,

$$N(z-\beta)\le c(z)\le N(z+\beta). \tag{6}$$

Every division is defined. Pivot sizes and quotient numerators have $O(q)$ bits, apart from the fixed input exponent cost. A count costs $O(n)$ operations. Although adjacent counts need not be monotone, (6) gives $c(x)\le c(y)$ whenever $y-x\ge 2\beta$.

For $H=2I+G$, $0\le G\le I$, counts at the endpoints certify the interval $[1/2,4)$. To locate descending rank j (one based), target ascending rank $n+1-j$. At a midpoint z, update the lower endpoint to $\max(\mathrm{lo},z-\beta)$ if $c(z)<n+1-j$; otherwise update the upper endpoint to $\min(\mathrm{hi},z+\beta)$. The exact eigenvalue remains in the closed bracket. Its width obeys $w_{\mathrm{new}}\le w/2+\beta+1/(2Q)$; after $2q+10$ steps it is at most $\eta^2/64$.

For the integer search define $c_g(Z)=c(Z/Q)$ and $\beta_g=Q\beta$. The integer coordinate Z represents the spectral coordinate Z/Q.

Lemma 3.1. *Suppose the integer interval $[L_g,R_g]$ has safe population upper bound $r_0=c_g(R_g+\beta_g)-c_g(L_g-\beta_g)$, and an integer $h_g\ge\max(1,4\beta_g)$ satisfies $R_g-L_g\ge(12r_0+4)h_g$. A deterministic search finds an integer center at distance at least h_g from the rescaled spectrum and its exact count in $O(\log(n+1))$ count calls.*

Proof. At the integer midpoint x_g, equality of $c_g(x_g-h_g-\beta_g)$ and $c_g(x_g+h_g+\beta_g)$ certifies the required interval. If they differ, discard $[x_g-2h_g,x_g+2h_g]$. The safe population bounds of the two remaining sides have sum at most r_0-1, by the separated-query ordering above. Choose the side with smaller population. Its length is at least $(R_g-L_g-1)/2-2h_g$, sufficient for the same length

condition with the new population. Each failure halves the population; population zero succeeds immediately. The midpoint rounding is absorbed by $h_g \geq 1$. □

Each application below uses $L_g = QL$, $R_g = QR$ and $h_g = Qh$; the chosen dyadics make these integers. Dividing the returned center and clearance by Q restores spectral units, with physical clearance h.

Take the largest dyadic $W \leq n\eta/4$, so $W > n\eta/8$. At each anchor aW choose a boundary in $[aW - W/4, aW + W/4]$ by Lemma 3.1, with the largest dyadic $h \leq W/[2(12n+4)]$. Here $h > \eta/512$ exceeds both $1/Q$ and 4β, and the length $W/2$ meets the rescaled lemma's length condition. Cache this deterministic choice by its integer anchor. Consecutive boundaries are between $W/2$ and $3W/2$ apart. A localized requested rank belongs, by the exact boundary counts, to one of three neighboring cells. At most three boundaries per request suffice. We call a cell a spectral packet. Distinct packets are disjoint, their halfwidths are at most $3n\eta/16$, and their endpoint clearance satisfies $h > \eta/512$. This proves the setup cost stated after Algorithm 1.

Applying the same search to G_R on $[\Delta^2, 2\Delta^2]$ with the largest dyadic $h \leq \Delta^2/(32n)$ gives

$$\Delta^2 \leq \theta \leq 2\Delta^2, \qquad \operatorname{dist}(\theta, \operatorname{spec} G_R) \geq \gamma = h > \Delta^2/(64n). \tag{7}$$

For $q = 3p + 32$, the length condition holds and $h > \Delta^2/(64n) \geq \epsilon^2/64 > \max(1/Q, 4\beta)$, so $Qh \geq \max(1, 4Q\beta)$. Both Grams share this cut and count. The gap in (7) is chosen by the algorithm, not imposed on the input singular values.

3.2 A positive rational approximation

For a packet (l, u) of H, let $\mu = (l+u)/2$, $a_0 = (u-l)/2$, $a = a_0 - h/2$ and initially $g = a_0 + h/2$. The two exterior neighboring ranks may be localized to increase g, while retaining $0 < a < g \leq 8$. Target eigenvalues satisfy $|\lambda - \mu| \leq a$ and exterior eigenvalues satisfy $|\lambda - \mu| \geq g$. Set

$$\ell = \frac{g-a}{g+a} \geq \frac{1}{512n}, \qquad y(x) = \frac{(x-\mu)^2 - ag}{(x-\mu)^2 + ag}.$$

The target has $y \leq -\ell$, and the exterior has $y \geq \ell$; $\sqrt{ag}\ell \geq h/4$. A whole-space packet needs only coordinate directions and requires no rational filter.

Starting with $r_0(y) = y$ and $\ell_0 = \ell$, define

$$\alpha_t = \ell_t^{-1/2}, \quad r_{t+1}(y) = \frac{2\alpha_t r_t(y)}{1 + \alpha_t^2 r_t(y)^2}, \quad \ell_{t+1} = \frac{2\sqrt{\ell_t}}{1+\ell_t}. \tag{8}$$

For $\ell \leq |y| \leq 1$, $\ell_t \leq \operatorname{sign}(y) r_t(y) \leq 1$. Indeed the scalar map has maximum 1 at $\sqrt{\ell_t}$ and equal endpoint minima ℓ_{t+1}. After $\lceil \log_2(q+3) \rceil$ steps $\ell_t \geq 1/2$. Subsequently, $z_t = (1-\ell_t)/(1+\ell_t)$ satisfies $z_{t+1} \leq z_t^2$ and $z_t \leq 1/3$.

Let $\delta = 2^{-20}\eta^6$ and

$$J_{\max} = \lceil \log_2(q+3) \rceil + \lceil \log_2(6q+27) \rceil \leq 19.$$

The iteration count is selected without a precision retry. With $Q_s = 2^{768+16q}$ and $L_0 = \lfloor Q_s \ell \rfloor$, use

$$L_{t+1} = \left\lfloor \frac{2\lfloor \sqrt{L_t Q_s} \rfloor Q_s}{Q_s + L_t} \right\rfloor \tag{9}$$

until $L_t/Q_s \geq 1 - \delta/16$. Each downward local defect is below $3/Q_s$. The map in (8) contracts logarithmic input errors by at most $1/2$. Its accumulated lower-bound defect is at most $16(J_{\max} +$

$1)/(Q_s\ell_{\min}) < \delta/64$, where $\ell_{\min} = 1/(512n)$. The ideal endpoint error is below $\delta/64$ by $J_{\max}$, so the stated stopping threshold is attained by then. For the exact packet projector Π, putting $\vartheta = \delta/4$ gives

$$F_0 = \frac{(I - r_J(y(H)))/2 + \vartheta I}{1 + 2\vartheta}, \qquad \delta I/8 \le F_0 \le I, \quad \|F_0 - \Pi\| \le \delta. \tag{10}$$

3.3 Coefficients without subtractive residue formulas

Write $\psi_\alpha(x) = 2\alpha x/(1 + \alpha^2 x^2)$. For $b, W > 0$ and $t = \sqrt{1 + b^2}$,

$$\begin{aligned} \frac{W\psi_\alpha(x)}{\psi_\alpha(x)^2 + b^2} &= \frac{W_- x}{x^2 + b_-^2} + \frac{W_+ x}{x^2 + b_+^2}, \\ b_- &= \frac{b}{\alpha(1+t)}, & b_+ &= \frac{1+t}{\alpha b}, \\ W_- &= \frac{W}{\alpha t(1+t)}, & W_+ &= \frac{W(1+t)}{\alpha t b^2}. \end{aligned} \tag{11}$$

Common denominators verify the identity. Starting with $(b, W) = (1/\alpha_{J-1}, 2/\alpha_{J-1})$ and transporting backwards gives the partial fractions of r_J. All magnitude operations are positive; no subtraction such as $1 - \alpha^2 b^2$ is evaluated. The full conjugate rank is $r = 2^{J+1} = O(q^2)$ and $r \le 2^{20}$; construction costs $O(r)$.

For completeness, positive addition is nonexpansive in the maximum logarithmic input error and a rounded real operation adds at most $2u$. The scale recurrence has logarithmic derivative at most $1/2$, giving scale and α errors at most $12u$ and $10u$. The inverse-branch logarithmic derivatives in b are $\pm 1/t$, and those for W have magnitude at most 3. After v levels, $e_b \le (12 + 20v)u$ and $e_W(v+1) \le e_W(v) + 3e_b(v) + 30u$. Therefore

$$\max\{\left|\log(\widehat{b}/b)\right|, \left|\log(\widehat{W}/W)\right|\} \le 128(J+1)^2 u. \tag{12}$$

The initial packet data are dyadics imported exactly before the controlled division. The logarithmic derivative bounds hold throughout the positive domain, so this is a finite perturbation estimate.

Each b gives poles

$$z = \mu \pm \frac{\sqrt{ag}}{\sqrt{1+b^2}} + i\frac{\sqrt{ag}\,b}{\sqrt{1+b^2}}$$

and their conjugates. The corresponding weights are $-W\sqrt{ag}(1 + ib)/[4(1 + 2\vartheta)(1 + b^2)^{3/2}]$ and the negative conjugate, paired with those poles. Real and imaginary components are formed separately; cancellation in the real center is bounded absolutely. Reverse transport gives $\min(b, 1/b) \ge \ell/3^J$: each reverse level loses at most 3α, while the product of the inverse scales with the starting $\sqrt{\ell_{J-1}}$ is at least ℓ. Also $r_J(1) = \sum W/(1 + b^2) \le 1$, so the ideal absolute weight mass is at most $\sqrt{ag}$. The ideal pole clearance is greater than $2^{-44}\eta$; rounded poles have clearance at least $2^{-48}\eta$. Conservatively the stored weight mass is at most 16, the constant magnitude at most 2 and $\max(1, \|H\| + |z|) \le 32$.

At $b_{\text{coef}} = 1024 + 16q$ bits, resolvent identities and (12) bound the common coefficient-operator error by $2^{136} u_{\text{coef}} \eta^{-2}$. The two inverse-clearance factors account for a pole perturbation. Rounding the raw tridiagonal contributes at most $2^{104} u_s \eta^{-2}$ with $u_s = 2^{-(768+16q)}$. The wider core imports the stored dyadic coefficients exactly.

4 Joint columns from a positive structured filter

This section describes the exact column selection and structured operations whose finite arithmetic is treated in Section 5. Let $0 < F \le I$ approximate a packet projector of population s_0. For a selected coordinate set I_t, put

$$C_t = F[I_t, I_t], \quad R_t = F - F[:, I_t]C_t^{-1}F[I_t, :], \quad f_t = \operatorname{tr}(C_t^{-1}), \quad f_0 = 0.$$

An unselected coordinate j has shorted diagonal $d_j = (R_t)_{jj}$ and squared interpolation norm $h_j = \left\| C_t^{-1}F[I_t, j] \right\|^2$. Choose a coordinate whose score $(1 + h_j)/d_j$ is within twice the minimum, and normalize the column $R_t e_j$. Repeat only for the number of requested directions in the packet.

Lemma 4.1. *If* $\operatorname{tr} F \ge s_0 - 1/2$ *and* $t \le s_0$*, twice-minimum selection satisfies* $f_t \le 4n^3$*. If* $\|F - \Pi\| \le \delta_{\text{eff}}$ *with the smallness conditions below, the selected unit columns are close to a single orthogonal family in* $\operatorname{range}\Pi$*.*

Proof. Block inversion gives $f_{t+1} = f_t + \text{score}_{\text{chosen}}$. From $F^2 \le F$, the sum of the score numerators is at most $n - 2t + f_t$, and the sum of their denominators is at least $q_0 - t$, with $q_0 = s_0 - 1/2$. Thus

$$f_{t+1} \le f_t + \frac{2(n + f_t)}{q_0 - t}, \qquad f_t \le n\left[\frac{(q_0 + 1)(q_0 + 2)}{(q_0 - t + 1)(q_0 - t + 2)} - 1\right] \le 4n^3.$$

The formula follows by multiplying the recurrence for $n + f_t$. Every selected shorted diagonal is at least $1/(4n^3)$.

Set $H_t = \max(1, f_t)$. The corresponding selected block of Π has minimum eigenvalue at least $1/H_t - \delta_{\text{eff}}$. Partition $\Pi = \left[\begin{smallmatrix} C & D^T \\ D & E \end{smallmatrix}\right]$. Since $\Pi^2 = \Pi$, $D^T D = C - C^2$, and $X = [-C^{-1}D^T; I]$ satisfies $\|X\|^2 \le 1/\lambda_{\min}(C)$. Block elimination and a resolvent expansion of the selected inverse bound the shorting perturbation by

$$\frac{\delta_{\text{eff}} H_t}{1 - 2\delta_{\text{eff}} H_t} \le 2\delta_{\text{eff}} H_t.$$

For $\delta = 2^{-20}\eta^6$ and $\delta_{\text{eff}} \le 2\delta$, the chosen ideal remaining projector diagonal is at least $1/(8n^3)$. The normalization inequality $\|a/\|a\| - b/\|b\|\| \le 2\|a - b\|/\|b\|$ gives unit-column distance at most $128\delta n^{9/2} \le n\eta/4096$. Successive shortenings of the same projector produce orthogonal ideal columns. Projectors of disjoint packets are orthogonal as well. □

4.1 Semiseparable recurrences

A tridiagonal resolvent has a semiseparable representation. After combining the conjugate poles, store scalars a_i, column generators $u_i, v_i \in \mathbb{C}^r$ and diagonal propagators $P_i \in \mathbb{C}^{r\times r}$ such that

$$F_{ij} = u_i^T P_i \cdots P_{j+1} v_j \quad (i > j), \qquad F_{ji} = F_{ij}. \tag{13}$$

We will ensure $\|P_i\| \le 1$. All contractions here are bilinear transposes, as required by a complex symmetric resolvent; pairing yields a real symmetric F.

For a fixed selected set, add z to all selected diagonal entries at once. A prime denotes the derivative at $z = 0$. The forward state $S \in \mathbb{C}^{r\times r}$ and S' start at zero before the first site. At site i, first replace S by $P_i S P_i^T$ (omitting the initial propagation); the scalar $a_i(z) = a_i + z\mathbf{1}_{i\in I_t}$ is the shifted diagonal. The resulting state gives

$$d_i = a_i - u_i^T S u_i, \quad b_i = v_i - S u_i, \quad w_i = b_i/d_i, \quad S_{\text{new}} = S + b_i b_i^T/d_i \tag{14}$$

at a selected site; an unselected site does not update S. Differentiate all these expressions to propagate S'. The backward state $\mathsf{B} \in \mathbb{C}^{r\times r}$ and its derivative start at zero beyond the last site and satisfy

$$\mathsf{B}_{\text{new}} = (I - u_i w_i^T)\mathsf{B}(I - w_i u_i^T) + u_i u_i^T/d_i \tag{15}$$

at a selected site, followed by the appropriate P_i propagation. Differentiate this expression for B'. At an unselected site,

$$d_j^{\text{short}} = d_j - b_j^T \mathsf{B} b_j, \qquad h_j = d_j' - 2(b_j')^T \mathsf{B} b_j - b_j^T \mathsf{B}' b_j.$$

These give all scores with $O(nr^2)$ work and $O(nr + r^2)$ storage. The reverse derivative involves blocks of the inverse square of the whole selected matrix, rather than squares of isolated inverse blocks.

The stored d_i, w_i also give streaming forward and reverse triangular solves for $C_t^{-1}F[I_t, j]$. Multiplication by F uses two length-r states: one forward state accumulates the strictly lower triangle of (13) and one reverse state the strictly upper triangle. Both the solve and matvec cost $O(nr)$. They produce $R_t e_j$ without storing a dense F or R_t.

5 Finite arithmetic and the inner Gram guarantee

We now justify that the actual coefficients, generator supply, score scans and column computations satisfy the hypotheses just used. All bounds are conservative. The inner range needed for Algorithm 1 is $80 \le q \le 191$; the estimates below hold for $16 \le q \le 244$ when $n\eta \le 1/64$.

5.1 One common nearby filter

Within this subsection write $d_i = T_{ii}$ and $e_i = T_{i,i+1}$ for the entries of the rounded real symmetric tridiagonal T. Write its stored rational filter as

$$F_{\text{coef}} = fI + \sum_{\nu=1}^{r} \omega_\nu (T - z_\nu I)^{-1}.$$

Here f is the real constant, z_ν the nonreal poles and ω_ν their weights, paired by conjugation as constructed above. For a single pole z and weight ω, let $\gamma_z = |\Im z|$ and $L_z \ge \max(1, \|T\| + |z|)$. The computed left and right pivots are

$$p_i = d_i - z - e_{i-1}^2/p_{i-1}, \qquad q_i = d_i - z - e_i^2/q_{i+1}, \tag{16}$$

with endpoint terms omitted. The inverse diagonal is computed from the two-sided twist, $g_i = 1/(p_i - e_i^2/q_{i+1})$. Complex multiplication and division, expanded into real primitives, have normwise errors at most $16u\,|xy|$ and $16u\,|x/y|$ for $u \le 1/8192$; complex addition/subtraction has error at most $2u(|x| + |y|)$. The positive division denominator is $(\Re y)^2 + (\Im y)^2$. No relative componentwise cancellation bound is used.

For an upper-half-plane pole the pivot imaginary recurrences subtract nonnegative terms from $-\gamma_z$. Their magnitudes stay at least $\gamma_z/2$. Each local pivot defect is at most $d_0 = 128uL_z^2/\gamma_z$. Thus all left pivots are exact pivots of $A_L = T - zI + \operatorname{diag}(e_L)$ with $|e_L| \le d_0$, and similarly on the right. If $u \le \gamma_z^2/(8192L_z^2)$, then

$$-\Im A_L \ge \gamma_z I/2, \quad \|A_L\| \le 2L_z, \quad \left\|A_L^{-1}\right\| \le 2/\gamma_z.$$

The identity $\Im(A_L^{-1}) = -A_L^{-*}(\Im A_L)A_L^{-1}$ gives $\left|(A_L^{-1})_{ii}\right| \geq \gamma_z/(8L_z^2)$. Consequently every contiguous product of left propagations is at most $P_0 = 16L_z^2/\gamma_z^2$. Comparing the mixed left/right twist matrix first with A_L and then with $T - zI$ gives

$$\left|\widehat{g}_i - (A_L^{-1})_{ii}\right| \leq 4096uL_z^2/\gamma_z^3, \qquad |\widehat{g}_i| \leq 3/\gamma_z. \tag{17}$$

Zero offdiagonal entries give zero propagation and require no division by an offdiagonal entry. Conjugate channels remain exact conjugates under symmetric real rounding.

For raw propagation $\widehat{\phi}_{i+1} = \mathrm{fl}(-e_i/\widehat{p}_i)$, form positive scales $\widehat{s}_1 = 1$ and

$$\widehat{s}_i = \mathrm{fl}\left(\sqrt{\mathrm{fl}(1 + \mathrm{fl}((\mathrm{fl}(\left|\widehat{\phi}_i\right| \widehat{s}_{i-1}))^2))}(1 + 128u)\right).$$

Store $\widehat{U}_i = \mathrm{fl}(\mathrm{fl}(\omega\widehat{g}_i)\widehat{s}_i)$, $\widehat{V}_i = \mathrm{fl}(1/\widehat{s}_i)$ and $\widehat{P}_i = \mathrm{fl}(\widehat{\phi}_i\,\mathrm{fl}(\widehat{s}_{i-1}/\widehat{s}_i))$. The scales cancel algebraically in represented offdiagonal products. Per-edge relative errors are at most $128u$, and complete product errors are at most $1024nu$ for $u \leq 1/(8192n)$. The outward factor dominates the scale and ratio errors below $16u$, so $\left|\widehat{P}_i\right| \leq 1$. Expanding the positive scale recurrence using P_0 gives

$$1 \leq \widehat{s}_i \leq 4\sqrt{n}P_0, \qquad \left|\widehat{U}_i\right| \leq 512\,|\omega|\,\sqrt{n}L_z^2/\gamma_z^3.$$

The diagonal is accumulated separately and is not identified with an exact sum of the stored scaled generators.

For r paired channels and $u \leq 1/(1024r)$, the actually represented real symmetric filter satisfies

$$\|F_{\mathrm{sup}} - F_{\mathrm{coef}}\| \leq uC_{\mathrm{sup}}, \qquad C_{\mathrm{sup}} = 2^{20}n^2\sum_z \frac{|\omega_z|\,L_z^4}{\gamma_z^5} + 2^{10}r\left(|f| + \sum_z \frac{|\omega_z|}{\gamma_z}\right). \tag{18}$$

Indeed an offdiagonal entry differs from $\omega(A_L^{-1})_{ij}$ by at most

$$|\omega|\,P_0\left(4096uL_z^2/\gamma_z^3 + (3/\gamma_z)1024nu\right) \leq 2^{17}\,|\omega|\,nuL_z^4/\gamma_z^5.$$

Resolvent comparison with $T - zI$ and a maximum-row-sum bound give the first term in (18). The rounded diagonal accumulation adds at most $2^8ru(|f| + \sum|\omega|/\gamma_z)$, covered by the second term. Every channel is compared with the same original T. Together with the coefficient and input errors this gives one filter for every selected column.

5.2 Segment bounds for the state scans

Let an actually supplied F satisfy $cI \leq F \leq I$ and (13) with contractive P_i. Let U, V bound $\|u_i\|, \|v_i\|$ at every site and set

$$M = \max(2, n, r, c^{-1}, U, V), \quad h = c/2, \quad K = 1 + 2nUV/c.$$

Use the jet norm $\|(X, X')\|_h = \|X\| + h\,\|X'\|$, which is submultiplicative. Exact states are compressions of inverses of selected principal matrices. For example, at a cut after site a, let I_- be the earlier selected coordinates and $C_- = F[I_-, I_-]$. For $P_{t:s} = P_t \cdots P_s$ (the identity for an empty product), let the row for $j \in I_-$ of W_- be $v_j^T P_{a:j+1}^T$. Then $S(0) = W_-^T C_-^{-1} W_-$ and $S'(0) = -W_-^T C_-^{-2} W_-$, with $\|W_-\| \leq \sqrt{n}V$. On $|z| \leq h$ the inverse norms are at most $2/c$, so exact state jets are at most $3M^4/2$.

For the segment from after site a to after site b, let I_0 contain its m selected coordinates, $C_0 = F[I_0, I_0]$ and $P = P_{b:a+1}$. At nonzero z, the two selected blocks are $C_- + zI$ and $C_0 + zI$; the formulas below use these shifted blocks when evaluated on the disk. The ith rows, indexed by $i \in I_0$, of the $m \times r$ coupling matrices L_0, W_0 are respectively $u_i^T P_{i:a+1}$ and $v_i^T P_{b:i+1}^T$. Thus $\|L_0\| \le \sqrt{n}U$ and $\|W_0\| \le \sqrt{n}V$. Shorting the earlier coordinates changes the selected segment block to $C_0 - L_0 S L_0^T$ and its outgoing coupling to $W_0 - L_0 S P^T$. Eliminating this selected block, and adding the propagated incoming state, gives the complete-segment map

$$\Phi(S) = PSP^T + (W_0 - L_0 S P^T)^T (C_0 - L_0 S L_0^T)^{-1} (W_0 - L_0 S P^T).$$

Writing $D = C_0 - L_0 S L_0^T$, $Z = W_0 - L_0 S P^T$, $T_0 = P - Z^T D^{-1} L_0$ and $Q_0 = L_0^T D^{-1} L_0$, the finite Woodbury identity gives

$$\Phi(S + E) - \Phi(S) = T_0 E (I - Q_0 E)^{-1} T_0^T. \tag{19}$$

On the disk, $\|T_0\| \le K^2$ and $\|Q_0\| \le 2nU^2/c$. For $\|E\| \le r_0 = c/[8n \max(1, U^2)]$ the segment Lipschitz constant is less than $2K^4$, uniformly in the segment length.

At each actual computed input define the local defect against the exact one-step jet map. Replace it by the analytic function $E_i + zE_i'$. The resulting exact perturbed recurrence has the actual value and derivative at zero. Put $D_F = \sum_i (\|E_i\| + h\,\|E_i'\|)$. If $2K^4 D_F \le r_0/2$, both the state before injecting a defect and the state after it remain in the allowed neighborhood. Induct on the number of injections: telescope each earlier defect through its exact remaining segment, whose endpoints have already been shown safe. Equation (19) then gives disk error at most $2K^4 D_F$.

The recurrence is rational and has no pole on an open neighborhood of this closed disk. Cauchy's formula [5, Theorem 16.26] gives derivative error at most the disk error divided by h. Applying the scalar formula to $a^* E(z) b$ for arbitrary unit a, b avoids a dimension factor. Total jet error is at most $4K^4 D_F$. This induction also proves existence of the perturbed recurrences, rather than assuming it from the exact state bounds.

For the reverse scan, exact factors give error at most $2K^4 D_B$. Put $\Delta w_i = \|\widehat{w}_i - w_i\|_h$, $\Delta v_i = \left\|\widehat{d}_i^{-1} - d_i^{-1}\right\|_h$, $A_i = U \Delta w_i$ and $a_i = \left\|I - u_i w_i^T\right\|_h$. Using (15), the additional defect due to the forward factors is bounded by

$$(2a_i + A_i) A_i \left\|\widehat{\mathsf{B}}_i\right\|_h + U^2 \Delta v_i.$$

Let $\Lambda = \sum_i (2a_i + A_i) A_i$ and let E_B sum the reverse rounding defects and $U^2 \Delta v_i$. Exact reverse jets are at most $M_B = 2nU^2/c$. If $2K^4 \Lambda \le 1/2$, absorption gives reverse error at most $4K^4(E_B + \Lambda M_B)$. A first-exit argument closes the local state bounds used to estimate these defects.

5.3 Explicit rounding budgets

Assume $u \le 2^{-128} M^{-160}$. Expand complex operations into real ones, apply the geometric γ_{8r} summation bound, and use $\||X|\| \le \sqrt{r}\,\|X\|$. A dot or matrix-vector jet product has normwise roundoff factor at most $2^{10} u M^2$. Substitution into (14)–(15), at the same actual input, gives

Local quantity	Absolute jet defect
b	$2^{14} u M^7$
d	$2^{15} u M^8$
d^{-1}	$2^{26} u M^{17}$
one forward update	$2^{40} u M^{40}$
one reverse update with fixed factors	$2^{40} u M^{40}$

For the reciprocal, exact and computed denominator magnitudes are at least $1/(2M)$ and $1/(4M)$, and reciprocal-jet sensitivity is at most $2^{10}M^9$. The bounds on b, d, d^{-1}, w are respectively $4M^5, 4M^6, 18M^8, 72M^{13}$. The largest forward term is the reciprocal defect times bb^T, at most $2^{30}uM^{27}$. In reverse, $y = \mathsf{B}w$, $w^T y + 1/d$ and the rank-two update have magnitudes at most $216M^{17}, 2^{14}M^{30}, 2^{16}M^{32}$, respectively; the larger table bounds cover them.

There are at most M sites, $K \le 3M^4$, and $4K^4 \le 2^9 M^{16}$. The segment estimate gives forward jet error at most $2^{49}uM^{57}$, and hence

$$\Delta d \le 2^{50}uM^{59}, \quad \Delta b \le 2^{50}uM^{58}, \quad \Delta(d^{-1}) \le 2^{62}uM^{69}, \quad \Delta w \le 2^{66}uM^{75}.$$

It follows that $\Lambda \le 2^{74}uM^{91}$ and $E_B \le 2^{64}uM^{72}$. Reverse jet error is at most $2^9M^{16}(2^{64}uM^{72} + 2^{75}uM^{95}) \le 2^{85}uM^{111}$. The assumed u makes the neighborhood and absorption conditions strict. At every unselected coordinate the final score quantities satisfy

$$\left|\widehat{d}_j - d_j\right|, \quad \left|\widehat{h}_j - h_j\right| \le 2^{96}uM^{128}. \tag{20}$$

Derivative extraction divides by $h \ge 1/(2M)$; its largest earlier term is $\|b\|_h^2$ times the reverse jet error and fits this bound. Since $d_j \ge c$ and $h_j \ge 0$, minimizing $(1 + \max(0, \widehat{h}_j))/\widehat{d}_j$ selects a true score within twice the minimum. If also $2^{96}uM^{128} < 2^{-250}$, the retained negative-$h$ diagnostic cannot fire. More explicitly, the implementation rejects a nonpositive selected factor $\widehat{d}_i$, discards nonpositive candidate diagonals and fails if no positive candidate remains, and rejects a candidate with $\widehat{h}_j < -2^{-250}$. The preceding bounds rule out these predicates at the proof precision.

5.4 Both solves and the actual column

The exact factors for $C = F[I, I]$ give $C = LDL^T$ with $c \le D_{ii} \le 1$ and $\|L\|, \|L^{-1}\| \le c^{-1/2}$. Indeed Schur complements preserve the bounds, $LDL^T \le I$ with $D \ge cI$ bounds L, and $C^{-1} = L^{-T}D^{-1}L^{-1}$ bounds L^{-1}. The two streaming solves are exactly $Ly = b_I$ and $L^T x = D^{-1}y$.

Let $\left\|\widehat{L} - L\right\| \le a_0$, $\left\|\widehat{D} - D\right\| \le b_0$, $t = c^{-1/2}$, $a_0 \le \sqrt{c}/2$, $b_0 \le c/2$ and $\rho_0 = a_0(2t + a_0) + b_0(t + a_0)^2 \le c/2$. Then $\left\|\widehat{L}\widehat{D}\widehat{L}^T - C\right\| \le \rho_0$ and its inverse norm is at most $2/c$. Multiplying the two actual triangular equations, with equation defects e_f, e_t and right-hand-side error e_b, gives

$$\left\|\widehat{x} - C^{-1}b\right\| \le \frac{2}{c}[e_b + e_f + (t + a_0)(1 + b_0)e_t] + \frac{2\rho_0}{c^2}\|b\|. \tag{21}$$

Let $W_s = \max_{i \in I} \|\widehat{w}_i\|$. The preceding factor estimates give $W_s \le 73M^{13}$, by the bound $72M^{13}$ on exact factors and their smaller computed perturbation. A local streaming vector update has error at most $48u(\|\text{state}\| + W_s|\widehat{y}_i|)$. Regarding $\widehat{y}$ as exact input and summing along the contractive propagations gives state norm at most $2W_s\sqrt{n}\,\|\widehat{y}\|$ and state error at most $256un^{3/2}W_s\|\widehat{y}\|$, provided $u \le 1/[4096(n + r + 1)]$. Final dot products give equation defect at most $\kappa(\|b\| + \|\widehat{y}\|)$, with $\kappa = 2^{10}u[1 + n(n + r + 1)UW_s]$. The solution term is absorbed using $\left\|\widehat{L}^{-1}\right\| \le 2t$ and $2t\kappa \le 1/4$. The reverse solve has the same estimate, with its rounded diagonal division included. The two-scan matvec satisfies $\|\mathrm{fl}(Fx) - Fx\| \le 2^{16}uM^6\|x\|$.

The factor estimates above give $a_0 \le 2^{66}uM^{77}$, $b_0 \le 2^{50}uM^{59}$ and $\rho_0 \le 2^{72}uM^{80}$. All solve conditions follow from the imposed u. For $\widehat{b} = \mathrm{fl}(Fe_j)$, the forward and reverse equation defects are at most $2^{26}uM^{19}$ and $2^{30}uM^{20}$, respectively. Equation (21) gives $\|\widehat{x} - C^{-1}F[I, j]\| \le 2^{78}uM^{84}$. After the second matvec and subtraction, with $z = R_t e_j$,

$$\|\widehat{z} - z\| \le 2^{80}uM^{84}, \qquad \|\widehat{q} - z/\|z\|\| \le 2^{90}uM^{96}. \tag{22}$$

The Schur complement is at least cI, so $\|z\| \geq c$ and the first error is below $c/2$. The real computed column is therefore nonzero. Taking real parts cannot increase its distance to z; normalization adds at most twice the column error divided by c, and norm/square-root/division roundoff contributes at most $8nu$, all covered by the second bound.

5.5 The resulting tridiagonal procedure

Choose

$$b_{\mathrm{coef}} = 1024 + 16q, \quad b_s = 768 + 16q, \quad L_M = 224 + 6q, \quad b_{\mathrm{core}} = 1024 + 160L_M. \tag{23}$$

The pole and mass bounds give $C_{\mathrm{sup}} \leq 2^{274}\eta^{-7}$: the two terms in (18) are at most $2^{272}\eta^{-7}$ and $2^{83}\eta^{-1}$. Coefficient, Gram and supply errors are each at most $\delta/256$. Consequently $\|F_{\mathrm{sup}} - F_0\| \leq 3\delta/256 < \delta/64$ and every actual propagation is contractive. For $F_{\mathrm{scan}} = F_{\mathrm{sup}}/2$ we may take

$$c = \delta/32, \quad U \leq 2^{164}\eta^{-7/2}, \quad V \leq 2^{10}, \quad M \leq 2^{224}\eta^{-6}.$$

The core budget in (23) then implies

$$u_{\mathrm{core}} \leq 2^{-128}M^{-160}, \quad 2^{96}u_{\mathrm{core}}M^{128} < 2^{-250}, \quad 2^{90}u_{\mathrm{core}}M^{96} \leq n\eta/4096.$$

Exact power-of-two scaling of the supplied diagonal and U scales every binary primitive and state accordingly: score minimizers, interpolation coefficients $C_t^{-1}F[I_t, j]$ and normalized columns agree. Thus this analysis applies to the unscaled implementation. For the exact selection comparison use $\overline{F} = F_{\mathrm{sup}}/(1 + \delta/64)$, which satisfies $0 < \overline{F} \leq I$ and $\left\|\overline{F} - \Pi\right\| < 2\delta$. An exact positive scalar changes every exact score by a common factor and preserves normalized shorted columns. Lemma 4.1 therefore applies to the same selected sequence.

Proposition 5.1. *For $0 \leq G \leq I$ real symmetric tridiagonal, $n\eta \leq 1/64$ and $16 \leq q \leq 244$, the specified procedure returns centers at the requested ordered ranks and jointly computed unit directions with simultaneous value, residual and pairwise inner-product errors at most $512n\eta$. It uses the work stated after Algorithm 1, without reorthogonalizing returned vectors.*

Proof. Equations (20)–(23) give twice-minimum true selection and numerical column errors at most $n\eta/4096$. Lemma 4.1 compares all selected columns with one orthogonal family, giving total internal direction error at most $n\eta/2048$. Packet centers have the correct ranks and halfwidth at most $3n\eta/16$. These bounds imply the stated larger value, residual and inner-product budget. The implementation exports 100 decimal digits only after selection and solves. Each exported component has error at most 10^{-99}; throughout $q \leq 244$, $\sqrt{n}10^{-99} < n\eta/4096$. The imported direction is normalized again. The finite iteration, rank and count bounds give the announced work. □

6 Recovery and proof of the main theorem

Recall $A = B/s$, $\Delta = n\epsilon$, $\xi = 2^{-20}n\epsilon^3$ from (5), with the common squared-spectrum cut θ and clearance γ in (7). Take $q = 3p + 32$, so $512n\eta = \xi/8$. Retaining recovery margin, the one joint right call supplies unit directions and centers with

$$|\alpha_i - \lambda_{j_i}(G_R)|, \quad \|(G_R - \alpha_i I)v_i\| \leq \xi, \qquad \left|v_i^T v_j\right| \leq \xi \quad (i \neq j).$$

The independent left call supplies the same bounds for w_i, β_i on only the requested near-zero indices. By (7), their squared values are at most $\theta - \gamma$, while positive ones are at least $\theta + \gamma$; also $\xi \leq \gamma/4$.

For a positive rank set $r_i = (G_R - \alpha_i I)v_i$, $t_i = \|Av_i\|$ and $u_i = Av_i/t_i$. Then $t_i^2 \geq \theta + \gamma - 2\xi \geq \Delta^2$. The first residual is zero; the second is $(I - v_i v_i^T)r_i/t_i$, of norm at most ξ/Δ. Also $\left|t_i^2 - \lambda_{j_i}(G_R)\right| \leq 2\xi$, so the singular-value error on the A scale is at most ξ/Δ. For two positive left directions,

$$u_i^T u_j = \frac{\alpha_j v_i^T v_j + v_i^T r_j}{t_i t_j}, \qquad \left|u_i^T u_j\right| \leq 3\xi/\Delta^2.$$

For a near-zero index output value zero and $u_i = w_i$. Its singular value and both residuals on the A scale are at most $\sqrt{2}\Delta$, since $\|Av_i\|^2, \left\|A^T w_i\right\|^2 \leq \theta - \gamma + 2\xi \leq 2\Delta^2$. No alignment between the two near-kernel bases is needed for zero values.

The cross inner products of left directions need a different estimate. Let v, α, r be positive and w, β, ℓ near zero, where $\ell = (G_L - \beta I)w$. The identity $G_L A = A G_R$ gives

$$(\alpha - \beta) w^T A v = \ell^T A v - w^T A r. \tag{24}$$

Here $t = \|Av\| \geq \Delta$, $\left\|A^T w\right\| \leq \sqrt{2}\Delta$ and $|\alpha - \beta| \geq 3\gamma/2$. Dividing (24) by t retains the near-kernel factor in its second term and yields

$$\left|w^T u\right| < 2\xi/\gamma < \epsilon/8192.$$

Replacing $\left\|A^T w\right\|$ by $\|A\|$ would lose a factor Δ^{-1}. The right directions already come from one joint call. Finally $3\xi/\Delta^2 = 3\epsilon/(2^{20}n)$ and $\xi/\Delta = 2^{-20}\epsilon^2$. Multiplication by $s \leq 4\|B\|$ restores the original value and residual scale.

For finite recovery keep the right direction at coefficient precision until Av_i, its norm and its division are complete. With $u_r \leq 2^{-b_{\text{coef}}}$, sequential fixed-precision norm sums and local bidiagonal products give errors at most $8nu_r$ for Av_i and $16nu_r$ for its norm. Reunitization contributes $O(nu_r)$, and the left direction error is at most $64nu_r/\Delta$. The choice $q = 3p + 32$ gives $64u_r/\epsilon < \epsilon/64$; all recovery errors fit between the $\xi/8$ inner budget and the ξ budget just used. No exact unbounded-length sum is used.

Final componentwise correct rounding changes a vector by at most ϵ times its pre-rounded norm. Mathematical unitization changes each represented direction by at most 3ϵ from the preceding direction and leaves the raw squared-norm defect at most 8ϵ. Nonnegative values receive at most 2ϵ additional error on the scaled matrix. Pairwise inner products increase by at most 6ϵ; the positive residuals are at most $\xi/\Delta + 10\epsilon$ and near-zero residuals at most $\sqrt{2}\Delta + 3\epsilon$ on the A scale. These estimates and $s \leq 4\|B\|$ prove (2)–(4) with constant 128. The zero-input branch is exact. The cost following Algorithm 1 includes all setup, recovery and output operations. Since $r = O(q^2)$ and $q = 3p + 32$, it proves the complexity clauses of Theorem 2.1.

Corollary 6.1. *For the raw returned arrays, ordered-value errors and both residual norms are bounded by $256n\epsilon\|B\|$, and separate pairwise inner products by $256n\epsilon$. Their squared-norm defects are at most 8ϵ; they are not asserted to have exact norm one.*

Proof. For a carrier $x = a\mathcal{U}(x)$, $|a - 1| \leq |a^2 - 1| \leq 8\epsilon$, so $\|x - \mathcal{U}(x)\| \leq 8\epsilon$. The value estimate and $128\Delta \leq 2$ give $0 \leq \widehat{\sigma} \leq 3\|B\|$. Each raw residual is therefore at most $(128\Delta + 32\epsilon)\|B\| \leq 160\Delta\|B\|$. For distinct carriers, their scale factors multiply the represented inner-product bound by at most $(1 + 8\epsilon)^2 < 2$. The unchanged value error and the norm estimate complete the result. □

7 Implementation and finite validation

The research library exposes one installed callable API, `selected_svd`, and a CLI using the same parameter resolver. Its problem inputs are the diagonal, superdiagonal and requested ranks; the public rank convention is zero-based descending. The default `auto` profile chooses exact coordinate vectors for zero input and the sufficient proof precision otherwise. An explicit `diagnostic` profile defaults to 4096 core bits and does not inherit the general precision guarantee. At $p = 53$, both profiles retain $q = 191$ and 4080 coefficient bits. The result records the effective precision, packet decisions and reasons. A missing requested backend raises an error without reducing precision.

The core expands complex operations over Boost `cpp_bin_float`; coefficient and controller arithmetic uses mpmath. Boost's documented basic arithmetic/square-root semantics [6] motivate the declared primitive model, but the implementation is not formally verified. Protected dimensions use signed 64-bit-capable indices with checked allocation products. Compiler, file, address-space, memory and time failures are operational boundaries.

An optional exact-instance checker reads the original input and output as rationals. Exact tridiagonal inertia bounds each requested ordered singular value. Sparse residual sums, raw norm defects and normalized inner products are checked by rational inequalities, using the stronger sufficient scale $\rho \le \|B\|$. This is a separate finite validation cost, not a free linear-work production step or a proof for untested inputs.

The two fixed validation matrices are

$$B_2 = \begin{bmatrix} 2 & 1/4 \\ 0 & 1 \end{bmatrix}, \qquad B_3 = \begin{bmatrix} 1 & 1/8 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

with all ranks requested. The first exercises coupled nonzero triplets; the second exercises a positive branch with a two-dimensional zero cluster. Default and explicit proof configurations are compared on identical inputs and output requirements. The diagnostic profile is reported separately because its general guarantee differs. NumPy/LAPACK singular values provide a finite independent comparison, while exact rational checks validate the original matrix's ordered values, both residuals, norms and left/right orthogonality. These cases do not establish asymptotic scaling or practical superiority.

Case	Profile	Core bits	Solve (s)	Exact check (s)
coupled2	auto	220224	23.728	0.0012
coupled2	proof	220224	23.905	0.0006
coupled2	diagnostic	4096	0.129	0.0007
rank_deficient3	auto	220224	42.427	0.0007
rank_deficient3	proof	220224	42.969	0.0008
rank_deficient3	diagnostic	4096	0.212	0.0007

Table 1: Fixed matched inputs at $p = 53$. All six exact instance checks passed. Times are individual observations, with build/cache costs recorded separately in the source distribution. Diagnostic precision has a different general guarantee.

The source distribution supplies tests, a fresh-install procedure, effective settings, fixed-case benchmark scripts and compact results. Build, solve and optional exact verification times are recorded separately. The accompanying curated source distribution contains the installation, test and reproduction instructions together with these fixed-case data. Its repository record identifies the immutable release revision.

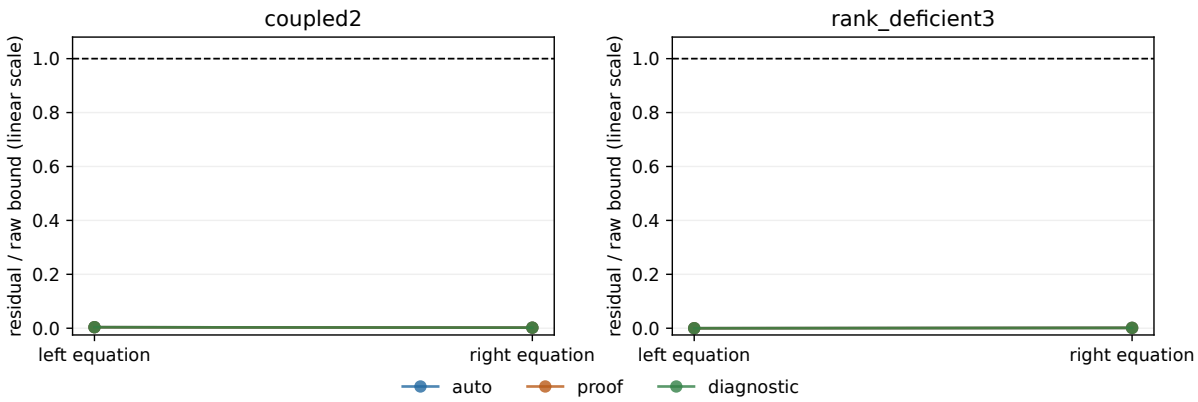


Figure 1: Raw-array residuals divided by $256n\epsilon\,\|B_2\|$ or $256n\epsilon\,\|B_3\|$, respectively. Smaller is better; the horizontal bound is 1. Zero observations are drawn at zero on a linear axis. The plot shows the fixed cases only; the three profiles coincide here. It does not establish a general error theorem.

AI assistance. The argument reconstruction, implementation, API, tests, benchmarks and writing were produced with AI-agent assistance under the author's direction. AI review and finite tests do not imply independent human review, formal proof certification or acceptance by a problem repository.

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